

CHAPTER THREE

HYPOTHESIS TESTING

In chapter two we have studied how to make estimations of the mean using point and interval estimations. The other aspect of statistical inference is known as statistical test of hypothesis. The branch of statistics which helps us in arriving at the criterion for deciding about the characteristics of the population, a parameter, based on the information obtained from the sample data is known as testing of hypothesis. We shall use the theoretical results presented for the interval estimation, and hence, a test of hypothesis is highly connected with the theory of estimation we studied before.

In this section, basically we will deal with testing hypotheses about population mean and population proportion. While doing so, we shall define some important terminologies which we may face and the errors we are committing in the process. We shall employ the standard normal distribution (or Z-test) and the t-distribution (or t-test), depending upon the nature of the population sampled and the sample.

3.1 Some terms in tests of hypothesis

Statistical hypothesis is defined as a statement (or an assertion) about the parameter of a population or its distribution that may be proved or disproved. Its plausibility is to be evaluated on the basis of information obtained by taking sample from the population.

Test statistic: is a statistic whose value serves to determine whether to reject or accept the hypothesis to be tested. It is a random variable.

A given statement concerning a parameter could be true or false. Hence we have two complementary hypotheses, namely, *null hypothesis and alternative hypothesis*.

a) Null hypothesis (H_0)

It is the hypothesis to be tested for possible rejection under the assumption that it is true and it is the hypothesis of equality or the hypothesis of no difference.

b) Alternative hypothesis (H_a)

It is hypothesis which is the complementary to the null hypothesis. It may be accepted if H_0 is rejected or be rejected if H_0 is accepted. It is the hypothesis of difference.

Alternative Hypothesis ~ three possibilities

- i. **Upper tailed test:** $H_A : \text{Population Characteristic} > \text{Hypothesized Value}$

ii. **Lower tailed test:** H_A : Population Characteristic $<$ Hypothesized Value

iii. **Two tailed test:** H_A : Population Characteristic $\neq$ Hypothesized Values

NOTE: The same Hypothesized value must be used in the Alternative Hypothesis as in the Null Hypothesis.

3.2 Steps in hypothesis testing

Step-1 The first step in hypothesis testing is to specify the null hypothesis (H_0) and the alternative hypothesis (H_1). Suppose the assumed or hypothesized value of μ is denoted by μ_o , then one can formulate two sided and one sided hypothesis as follows:

1. $H_0: \mu = \mu_o$ versus $H_1: \mu \neq \mu_o$ (two sided test)
2. $H_0: \mu = \mu_o$ versus $H_1: \mu < \mu_o$ (one sided test or left sided-test)
3. $H_0: \mu = \mu_o$ versus $H_1: \mu > \mu_o$ (one sided test or right sided-test)

Step-2: Specify a significance level of α .

Step-3 compute and select the test statistic.

The value of the test statistic can be calculated as follows:

The general formula of test statistic = $\frac{\text{sample statistic} - \text{hypothesized population parameter}}{\text{Standard error of the sample statistic}}$

Thus,

a) $Z_c = \frac{\bar{X} - \mu_o}{\frac{\sigma}{\sqrt{n}}}$ with known population SD.

b) $Z_c = \frac{\bar{X} - \mu_o}{\frac{S}{\sqrt{n}}}$ with unknown population SD

c) $t_c = \frac{\bar{X} - \mu_o}{\frac{S}{\sqrt{n}}}$ with unknown population SD and small sample size($n < 30$)

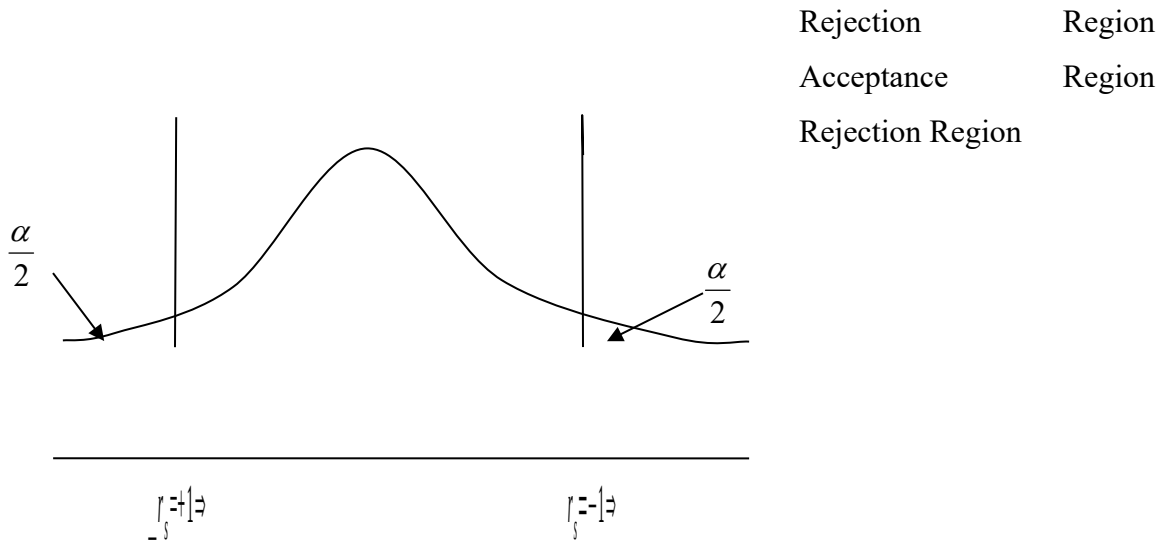
Where $\bar{X}$ is the sample mean and μ_o the parameter specified by the null hypothesis.

Step-4: formulate the decision rule.

a) For two sided test $H_0: \mu = \mu_o$ versus $H_1: \mu \neq \mu_o$, reject H_0 if

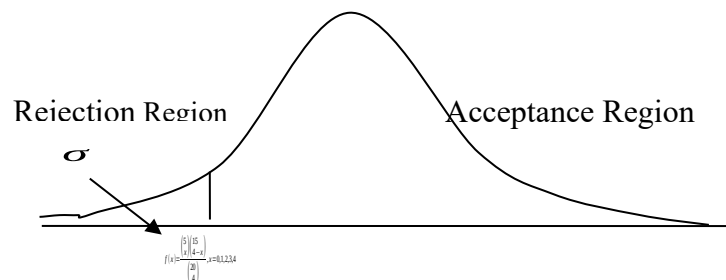
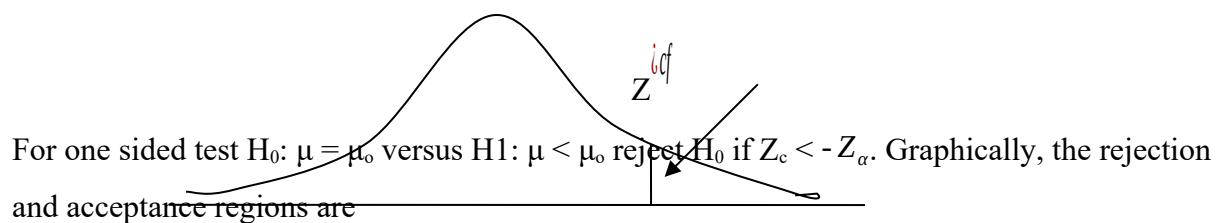
$$Z_c > Z_{\alpha/2} \text{ or } Z_c < -Z_{\alpha/2}.$$

Graphically, the rejection and acceptance regions are:



b) For one sided test $H_0: \mu = \mu_0$ versus $H_1: \mu > \mu_0$ reject H_0 if $Z_{\text{calculated}} > Z_{\alpha}$. Graphically, the rejection and acceptance regions are

Acceptance Region Rejection Region (z_{α})



Step 5: Make the decision rule

3.3 Type I and Type II errors

Statistical test of hypothesis can lead to two kinds of errors. If the statistical test rejects H_0 when it is true, the error is type I error. If the test accepts H_0 when it is false, the error is a type II error.

One could make two types of errors in Hypothesis testing. These are Type I and Type II errors.

The following table gives a summary of types of errors:

		The true situation is	
		H_0 is true	H_a is true
Action	Reject H_0 (Accept H_a)	Type I error	Correct decision
	Reject H_a (Accept H_0)	Correct decision	Type II error

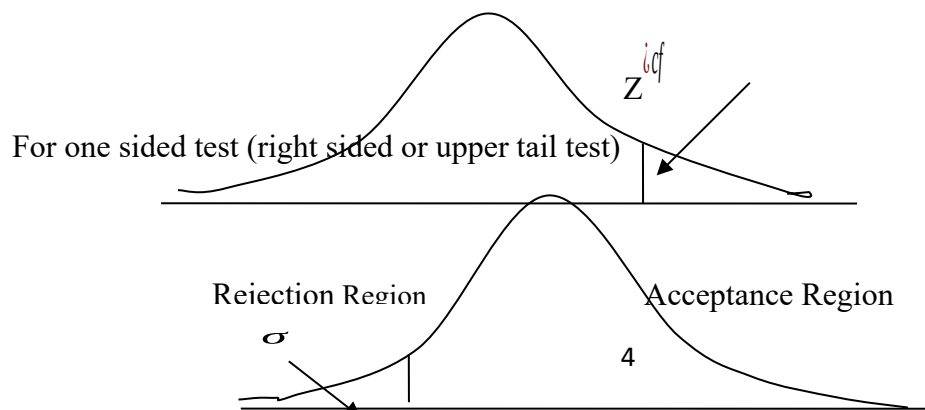
Type I error is the error committed in rejecting the null hypothesis when it is true.

Type II error is the error committed in accepting the null hypothesis when it is false. In both types of errors, a wrong decision has occurred. An ideal test procedure is one which is so planned as to safeguard against both these errors. However, in practical situations an attempt to minimize one of the errors maximizes the other. In view of this dilemma and the fact that wrong rejection of H_0 is a more serious error, we will hold $\sum_{d=0}$ at a predetermined low level, such as 0.1, 0.05, or 0.01 when choosing a rejection region. The level of significance 5% $\alpha = 0.05$ implies that in 5 samples out of 100 we are likely to reject a correct H_0 . In other words this implies that we are 95% confident that our decision to reject H_0 is correct.

3.4 One tailed and two tailed hypothesis test

Directional tests, or **one-tailed tests**, are hypothesis tests where the alternative hypothesis is stated as greater than ($>$) or less than ($<$) a value stated in the null hypothesis. Hence, the researcher is interested in a specific alternative from the null hypothesis. **A test is one tailed when the alternative hypothesis (H_a) states a direction.**

Acceptance Region Rejection Region (α)

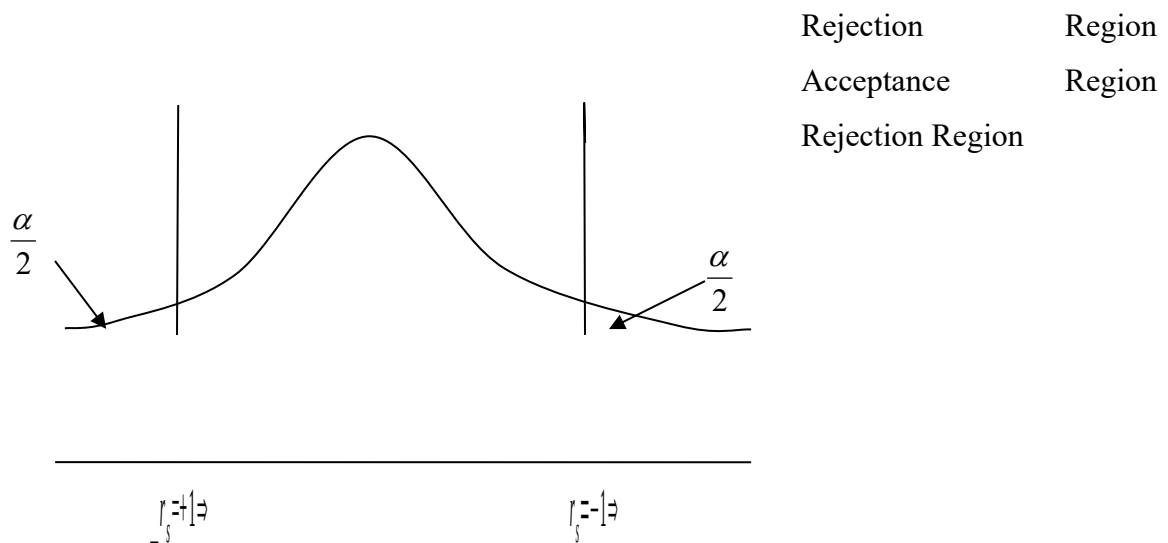


$$f(x) = \frac{\binom{n}{x} \left(\frac{\alpha}{n}\right)^x \left(1 - \frac{\alpha}{n}\right)^{n-x}}{\binom{n}{x}}$$

For one sided test (left sided or lower tail test)

Non directional tests, or **two-tailed tests**, are hypothesis tests where the alternative hypothesis is stated as *not equal to* ($\neq$). The researcher is interested in any alternative from the null hypothesis. **In two tail test we are concerned whether there had been a change in the population parameter.**

The critical values for a one tailed test are different from a two tailed test of the same level of significance. In two tailed test we **split the significance level in half in the lower tail and half in the upper tail.**



3.5 Hypothesis testing of a population means and proportion

Hypothesis testing of a population mean

Example2: The mean life time of a sample of 16 fluorescent light bulbs produced by a company is computed to be 1570 hours. The population standard deviation is 120 hours. Suppose the hypothesized value for the population mean is 1600 hours. Can we conclude that the life time of light bulbs is decreasing?

(Use $\alpha = 0.05$ and assume the normality of the population)

Solution:

Let μ = Population mean. , $\mu_0 = 1600$

Step 1: Identify the appropriate hypothesis

$$H_0 : \mu = 1600 \quad \text{vs} \quad H_1 : \mu < 1600$$

Step 2: select the level of significance, $\alpha = 0.05$ (given)

Step 3: Select an appropriate test statistics

Z- Statistic is appropriate because population variance is known.

Step 4: identify the critical region.

$$\text{The critical region is } Z_{\text{cal}} < -Z_{0.05} = -1.645$$

$$\Rightarrow (-1.645, \infty) \text{ is acceptance region.}$$

Step 5: Computations:

$$Z_{\text{cal}} = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} = \frac{1570 - 1600}{120 / \sqrt{16}} = -1.0$$

Step 6: Decision

Accept H_0 , since Z_{cal} is in the acceptance region.

Step 7: Conclusion

At 5% level of significance, we have no evidence to say that the life time of light bulbs is decreasing, based on the given sample data.

Example 2: Test at $A = \bar{x}$ whether the mean of a random sample of size $n = 16$ is "significantly less than 10" if the distribution from which the sample was taken is normal,

$$\frac{y_1 y_2 + y_2 y_3 + \dots + y_{n-1} y_n}{y_1^2 + y_2^2 + \dots + y_n^2} \quad \text{and} \quad \frac{y_1 y_2 + y_2 y_3 + \dots + y_{n-1} y_n}{y_1^2 + y_2^2 + \dots + y_n^2} \quad (\text{known}).$$

Solution:

$$* \quad G.M = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n} \quad \text{versus,} \quad GM = \sqrt{3 \times 12} = \sqrt{36} = 6$$

$$* \quad H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} \quad (\text{critical value})$$

$$* \quad H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} \quad (\text{calculated value})$$

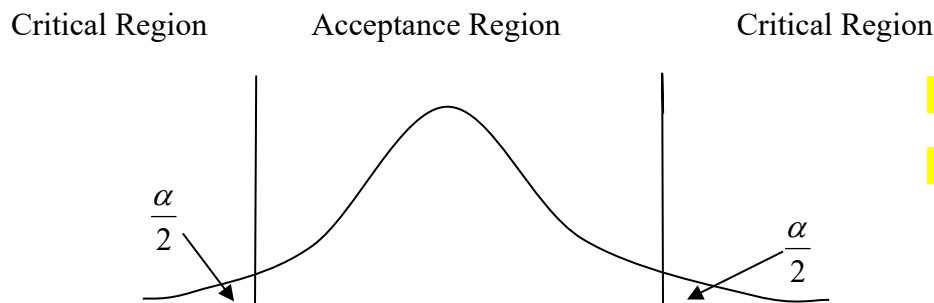
* Since $H.M = \frac{n}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_m}{x_m}}$, the null hypothesis is rejected. That is, the population means 8.4 is significantly less than 10 at 5% level of significance.

Example 3: Assume that in a certain district the mean systolic blood pressure of persons aged 20 to 40 is 130 mm Hg with a standard deviation of 10 mm Hg. A random sample of 64 persons aged 20 to 40 from village x of the same district has a mean systolic blood pressure of 132 mm Hg. Does the mean systolic blood pressure of the dwellers of the village (aged 20 to 40) differ from that of the inhabitants of the district (aged 20 to 40) in general, at a 5% level of significance?

$$\text{Solution:} \quad n = \sum_{k=1}^m f_k \quad n = \sum_{k=1}^m f_k \quad \text{Median} = L + \frac{\frac{n}{2} - CF}{f_{med}} \quad \text{and hence } f_{med}$$

$$Z_{0.025} = 1.96$$

$$CF = f_{med}$$



$$\bar{X} = -1.96 \quad 0 \quad 1.96$$

Since Z_c is in the acceptance region, $\mu_1 = \mu_0$ is accepted. That is, the systolic blood pressure of persons (aged 20 to 40) living in village x is the same as the mean systolic blood pressure of the inhabitants (aged 20 to 40) of the district.

Example 4: A sample of 16 students gave an average mark of 53.8 with a standard deviation of 5.2. Can the population means of marks is 50 at $\alpha = 0.05$?

Solution: $\alpha = 0.05$ $\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times w$

$x_1, x_2, \dots, x_n$ and hence (Q_i)

$$Q_i = L + \frac{w}{f_{Q_i}} \left(\frac{in}{4} - CF \right)$$

$$X_1, X_2, \dots, X_n$$

Since $X_1, X_2, \dots, X_n$ is rejected. i.e the population mean mark is significantly different from 50 at $\frac{L - S}{L + S}$

Example3: It is known in a pharmacological experiment that rats fed with a particular diet over a certain period gain an average of 40 gms in weight. A new diet was tried on a sample of 20 rats yielding a weight gain of 43 gms with variance 7 gms². Test the hypothesis that the new diet is an improvement assuming normality.

Solution: $\mu_0 = 40, \bar{X} = 43, n = 20, S = \sqrt{7}, \alpha = 0.05$

1 $H_0: \mu = 40$ vs $H_1: \mu > 40$

2 $\alpha = 0.05$

3 Critical Region: $t > t_{0.05}(19) = 1.729$

4 Computations: $t_{\text{cal}} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{43 - 40}{\sqrt{\frac{7}{20}}} = 5.07$

5 Conclusion: Reject H_0 and conclude that the new diet is an improvement.

Hypothesis testing for population proportion

Hypothesis testing for population proportion P is carried out in the same way as hypothesis testing for population mean when large samples and normality assumptions are fulfilled.

The test statistic is:

$$Z_c = \frac{\hat{P} - p_o}{\sqrt{\frac{p_o q_o}{n}}} \sim N(0, 1) \text{ where } q_o = 1 - p_o$$

The decision rule is:

a) For two sided test $H_o: P = P_o$ versus $H_1: P \neq P_o$ reject H_o if

$$Z_c > Z_{\alpha/2} \text{ or } Z_{\text{calculated}} < -Z_{\alpha/2}.$$

b) For one sided test $H_o: P = P_o$ versus $H_1: P > P_o$ reject H_o if

$$Z_c > Z_{\alpha}$$

c) For one sided test $H_o: P = P_o$ versus $H_1: P < P_o$, reject H_o if

$$Z_c < -Z_{\alpha}.$$

Example 1: A pharmaceutical company claims that a drug which it manufactures relieves cold symptoms for a period of 10 hrs in 90% of those who take it. In random sample of 400 people with colds who take the drug, 350 find relief for 10hrs. At a 0.05 level of significance, is the manufacture's correct?

Solution: $\frac{350}{400} = 0.875$, $\frac{350}{400} = 0.9$, $\alpha = 0.05$, $n = 400$, $Z_c = -1.645$

$H_o: P = 0.90$ Vs $H_1: P < 0.90$

Using the z statistic, we have

$$Z_c = \frac{\hat{P} - p_o}{\sqrt{p_o(1 - p_o)/n}} = \frac{0.875 - 0.90}{\sqrt{0.90(0.10)/400}} = -1.67$$

Since computed value of $Z_c = -1.67$ is less than the critical value of $-Z_{0.05} = -1.645$, therefore, the null hypothesis is rejected. The manufacture's claim is not upheld.